
ESP-YOLO: Stabilizing Incremental Object Detection via Eigenvalue Scaled Projection Regularization and Confidence-Aware Distillation

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Abstract

1 Incremental Object Detection (IOD) requires sequentially learning to localize and
2 classify novel object categories while preserving the stability of historical rep-
3 resentations to mitigate catastrophic forgetting. While single-stage frameworks
4 like YOLO are ideal baselines for real-world IOD deployment due to their ef-
5 ficiency, they remain under-explored. This gap primarily stems from YOLO’s
6 end-to-end prediction mechanism, where backpropagated gradients from new tasks
7 cause drastic perturbations to backbone parameters. Unlike two-stage models
8 that inherently isolate such impacts via class-agnostic region proposal networks,
9 YOLO suffers from significantly more severe intermediate feature drift and exacer-
10 bated class confusion. To overcome these architectural bottlenecks, we propose a
11 framework featuring Eigenvalue Scaled Projection Regularization (ESPReg) and
12 Confidence-Aware Distillation (CAD). Specifically, ESPReg utilizes the eigenval-
13 ues of feature covariance to quantify the importance of spatial directions, guiding
14 weight updates to minimize interference with old tasks. We theoretically prove
15 that ESPReg effectively maintains intermediate feature stability. Furthermore,
16 CAD leverages high-confidence regions to guide the distillation process, isolating
17 background interference to resolve severe class confusion. Extensive experiments
18 across single-stage and multi-stage settings demonstrate that our approach sur-
19 passes state-of-the-art IOD methods by 1.4% and 7.3% on the PASCAL VOC and
20 MS COCO benchmarks, respectively.

21 1 Introduction

22 Unlike human cognition, which can accumulate knowledge incrementally, deep neural networks
23 typically suffer from catastrophic forgetting when exposed to non-stationary data distributions.
24 Consequently, Incremental Learning (IL) [27] is proposed to enable models to sequentially learn new
25 tasks without compromising the proficiency acquired from historical data. While IL has garnered
26 widespread attention in image classification, Incremental Object Detection (IOD) holds significantly
27 greater practical application value, standing as a critical yet challenging frontier. Unlike classification,
28 object detection requires the model to simultaneously categorize and localize objects of interest.
29 Consequently, IOD must preserve both the discriminative knowledge and the precise localization
30 capabilities for historical categories, which imposes more stringent requirements on preserving model
31 stability for previously learned tasks.

32 Recent studies [5, 29, 32, 38] find that by projecting the gradients of weight to the null space
33 of previous tasks’ features, models can preserve their performance on previous trained datasets
34 after being trained on new dataset. This strategy is referred to as Null-Space Gradient Projection
35 (NSGP) [30]. However, there has been very limited attempts to apply NSGP to IOD. Only recently,
36 the state-of-the-art IOD method NSGP-RePRE [30] introduces NSGP into IOD setting. However,

existing methods naively apply NSGP to two-stage object detection models. Given that object detection is an inherently application-driven task, it imposes stringent demands on both efficiency and effectiveness. Consequently, single-stage frameworks like YOLO [19] serve as ideal baselines for real-world IOD. We identify two critical bottlenecks in applying NSGP to YOLO, as detailed below.

First, single-stage detectors suffer from severe intermediate feature drift. Unlike two-stage models (e.g., Faster R-CNN [20]) that inherently isolate the backbone from drastic semantic shifts via a class-agnostic Region Proposal Network (RPN), YOLO relies on an end-to-end dense prediction mechanism. Consequently, backpropagated gradients from novel tasks cause drastic perturbations to the backbone parameters. Traditional NSGP has been empirically proven to be less effective on YOLO-based incremental detectors.

Second, the lack of architectural decoupling in YOLO leads to cross-task class confusion. In a class-incremental setting, the ground truth frequencies of old and novel classes are naturally imbalanced. Therefore, when the YOLO-based detector is incrementally trained on both old and novel classes, it has a pronounced tendency to misclassify old classes into visually similar novel classes (eg. cow→sheep, bus→train). This leads to low-performance in task-agnostic inference scenario.

To overcome these architectural bottlenecks, we propose a novel framework, ESP-YOLO, to stabilize single-stage IOD via Eigenvalue Scaled Projection Regularization (ESPREg) and Confidence-Aware Distillation (CAD). ESPReg resolves the intermediate feature drift with an adaptive regularization loss. Our approach is grounded in the theoretical analysis of the feature drift risk, in which we identify that the risk’s upper bound is contributed by the projection of parameter updates onto each principal components of historical features, scaled by corresponding feature covariance eigenvalues. Based on this key insight, ESPReg replaces rigid gradient projection in NSGP [30] with a soft, adaptive regularization in loss function. Specifically, following the training of each task, we apply Principal Component Analysis (PCA) [18] to the input feature space. Subsequent parameter updates are projected onto these principal components, and the norms of these projections are weighted by their respective eigenvalues and summed to formulate a regularization term. By converting hard constraints into soft, eigenvalue-scaled energy barriers, ESPReg permits updates within the significant subspace when task-specific gradients outweigh the projection penalty. This mechanism effectively bounding weight deviation in critical directions to prevent feature drift.

Furthermore, minimizing feature drift through ESPReg alone is insufficient to fully resolve the challenges inherent in YOLO-based IOD. To address the challenge of cross-task class confusion, we introduce the Confidence-Aware Distillation (CAD) mechanism. Unlike standard approaches that indiscriminately distill all output channels, thereby risking the transfer of task-irrelevant background noise, CAD selectively transfers knowledge guided by the base model’s confidence. By generating a confidence mask based on the maximum classification scores across classes, CAD compels the model to anchor its predictions on regions where the base model exhibits high certainty. This targeted strategy effectively isolates background interference and preserves discriminative capabilities for historical categories, ensuring the stability of YOLO in IOD. The contributions of our work are summarized as:

- To address the severe intermediate feature drift in single-stage incremental detectors, we propose ESPReg. Grounded in a theoretical analysis of the feature drift risk, ESPReg adaptively penalizes parameter updates along critical historical feature directions based on their covariance eigenvalues. This mechanism effectively preserves structural stability while maintaining the necessary plasticity for learning novel tasks.
- To mitigate the cross-task class confusion in IOD, we introduce CAD. Instead of indiscriminate global distillation, CAD anchors the knowledge transfer exclusively on high-certainty regions, shielding the IOD process from background noise and ensuring semantic discrimination for incremental classes.
- Extensive experiments on Pascal VOC demonstrate the effectiveness of ESPReg across various baseline models. Specifically, it achieves a new state-of-the-art result on the 15+5 class incremental setting of Pascal VOC and 40+40 class incremental setting of MS COCO by 1.4% and 7.1% improvement over the NSGP-RePRE respectively.

2 Related Work

Incremental Learning (IL) aims to sequentially learn from novel data without catastrophically forgetting prior knowledge. In visual domains, methods are primarily evaluated under **class-incremental learning** (accommodating new categories) and **domain-incremental learning** (adapting to distribu-

tion shifts) [27]. Existing paradigms generally fall into three categories. **Regularization-based methods**, including distillation, constrain parameter updates to preserve crucial historical weights [11, 10]. However, they fundamentally struggle with the stability-plasticity dilemma: over-regularization hinders new learning (intransigence), while under-regularization fails to mitigate forgetting. Additionally, distillation variants degrade under severe distribution shifts; an overfitted teacher model yields unreliable predictions on novel inputs, thereby undermining its instructional efficacy. **Replay-based methods** alleviate distribution shifts by retaining or generating historical samples for joint training [24, 21]. Despite often achieving state-of-the-art performance, they incur substantial memory overhead and present privacy risks due to sensitive data storage. **Dynamic-architecture-based methods** minimize inter-task interference by allocating dedicated sub-networks or dynamically routing parameters [16, 28]. Nevertheless, their task-linear parameter growth makes them computationally prohibitive for resource-constrained deployments, and their frequent reliance on task identities during inference severely restricts their real-world applicability.

Incremental Object Detection (IOD). IOD introduces unique challenges beyond image classification, such as the coupled nature of localization and classification and the spatial sensitivity of features. The field was largely formalized by [25] who introduced a distillation-based paradigm. Subsequently, both distillation-based [25, 7, 15, 2, 35] and replay-based [23, 31, 14] methods have seen extensive study. However, the significant limitations of dominant paradigms persist: Given the efficiency demands of deployed object detection systems, dynamic architecture-based methods generally fall short because of its constantly expanding model size. Furthermore, replay-based strategies suffer from privacy concerns and memory burdens. Therefore, we direct our focus toward regularization-based methods, as they offer a more promising and practically viable paradigm for IOD.

Null-Space Gradient Projection (NSGP). NSGP [5, 22, 12, 36, 32, 38], a family of regularization-based methods, is steadily emerging as a promising paradigm. It mitigates forgetting by projecting gradients onto the null space of the historical features, theoretically ensuring zero interference to old tasks. Despite its success in classification and two-stage detection, the application of NSGP in YOLO is hindered by two critical bottlenecks, severe intermediate feature drift and class confusion. Departing from existing paradigms, we introduce a novel framework featuring ESPReg and CAD. It adaptively penalizes weight updates in critical feature sub-spaces and selectively distills semantic knowledge, thereby applying NSGP in single-stage IOD.

3 Preliminaries

In this section, we formulate the Incremental Object Detection (IOD) problem in Sec. 3.1. Next, in Sec. 3.2, we present empirical results demonstrating two major architectural bottlenecks of YOLO-based incremental detectors: feature drift and cross-task class confusion. Finally, we establish a theoretical analysis for the upper bound of feature drift, explicitly relating it to the eigenvalues of the feature covariance matrix (Sec. 3.3).

3.1 Problem Definition

We formulate IOD as a sequential learning process over a stream of datasets $\mathbb{D} = \{\mathcal{D}_1, \dots, \mathcal{D}_T\}$. At stage t , the dataset $\mathcal{D}_t = \{(\mathcal{X}_t, \mathcal{Y}_t)\}$ becomes available, where $\mathcal{X}_t$ denotes the images and $\mathcal{Y}_t$ contains annotations for the task-specific class set $\mathcal{C}_t$. Following the class-incremental setting, the class sets are mutually disjoint across stages, *i.e.*, $\mathcal{C}_i \cap \mathcal{C}_j = \emptyset, \forall i \neq j$. The primary objective is to evolve the detector parameters from Θ_{t-1} to Θ_t using only $\mathcal{D}_t$ for training, while preventing performance degradation on all accumulated classes $\mathcal{C}_{1:t} = \bigcup_{k=1}^t \mathcal{C}_k$.

3.2 Bottlenecks of YOLO Incremental Detector

Applying the YOLO architecture directly to IOD reveals significant challenges in mitigating catastrophic forgetting. As shown in Fig. 1, our empirical analysis highlights two major architectural bottlenecks of YOLO-based incremental detectors: feature drift and cross-task class confusion.

We first investigate the bottleneck of feature drift. In IOD, the stability of the intermediate features for previous tasks is paramount to mitigating catastrophic forgetting. A critical observation in our study is the disparity in feature drift between YOLO-based and Faster R-CNN based incremental detectors. To quantify this phenomenon, we employ the drift at the output layer of the feature extractor (*i.e.*, representations prior to the prediction heads) as a representative metric. As Fig. 1a indicates, Faster R-CNN exhibits a significantly lower degree of drift compared to YOLO detectors (the calculation details of feature drift are provided in Appendix A.1).

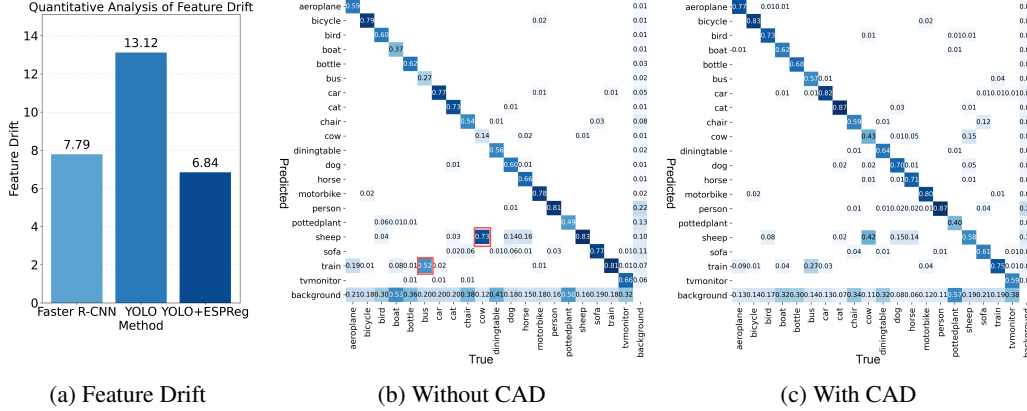


Figure 1: **Bottlenecks of YOLO-based IOD.** (a) The comparison of feature drift between Faster R-CNN, YOLO and YOLO+ESPReg, in which ESPReg is the method we propose in Sec. 4.1. The values in (b) and (c) represent the classification confidence. If the values on the diagonal are the highest, it indicates that the classification is correct. (b) When only ESPReg is applied, the classifier is biased by the current task’s data distribution, leading to severe misclassification (red box) of historical categories (Task-1) into visually similar new categories (Task-2). (c) With the Confidence-Aware Distillation (CAD), which is proposed in Sec. 4.2, cross-task confusion is mitigated.

This discrepancy is largely due to the end-to-end dense prediction paradigm of YOLO, which induces overwhelming back-propagated gradients that easily perturb historical knowledge embedded in the feature extractor parameters. In contrast, Faster R-CNN demonstrates superior feature stability because its class-agnostic Region Proposal Network requires minimal reconfiguration for new categories, naturally constraining back-propagated gradients and isolating the backbone from drastic shifts. Therefore, in response to the feature drift faced by the incremental learning of YOLO, we designed ESPReg in Sec. 4.1.

Fig. 1b demonstrates the second bottleneck of YOLO-based incremental detectors: cross-task class confusion. When sequentially trained on the Pascal VOC 15+5 class-incremental setting, YOLO exhibits a pronounced tendency to misclassify instances from the initial 15 classes into the 5 novel classes (e.g., cow→sheep, bus→train in the red box of Fig. 1b).

We attribute YOLO’s severe cross-task class confusion to its one-stage training paradigm. Under extreme incremental data imbalance, the BCE loss induces constant negative gradients that systematically suppress the output channels of historical tasks. This creates a critical confidence scale mismatch, overshadowing old categories during task-agnostic inference. Conversely, Faster R-CNN mitigates this via a Softmax-based classifier on sampled RoIs, which is tuned on the second stage. By coupling class probabilities, the logit growth of novel classes drives the negative gradients applied to historical channels toward zero, naturally preserving their relative posterior distributions. Therefore, we design a confidence-aware distillation mechanism to calibrate cross-task classification confidence.

3.3 Feature Drift Analysis

Feature drift is a key bottleneck affecting the incremental learning of YOLO. Therefore, we conducted a theoretical analysis of the feature drift. To establish a theoretical bound on the feature drift of YOLO-based incremental detectors, we analyze the influence of parameter updates on the network’s intermediate features. The detailed analysis process and formula derivation can be found in the appendix A.2. Without loss of generality, consider the l -th layer of the network with input features $\mathbf{x}^l \in \mathbb{R}^{d_{in}}$. Let $\mathbf{W}_{t-1}^l \in \mathbb{R}^{d_{out} \times d_{in}}$ denote the parameters converged at stage $t-1$. The forward propagation is given by:

$$\mathbf{x}^{l+1} = \sigma(\mathbf{W}_{t-1}^l \mathbf{x}^l), \quad (1)$$

where $\sigma(\cdot)$ is a non-linear activation function. When learning the new task at stage t , the weights are updated by a perturbation $\Delta \mathbf{W}_t^l$, resulting in $\mathbf{W}_t^l = \mathbf{W}_{t-1}^l + \Delta \mathbf{W}_t^l$. The resulting shift in the output feature is:

$$\Delta \mathbf{x}^{l+1} = \sigma((\mathbf{W}_{t-1}^l + \Delta \mathbf{W}_t^l) \mathbf{x}^l) - \sigma(\mathbf{W}_{t-1}^l \mathbf{x}^l). \quad (2)$$

To quantify forgetting, we examine the magnitude of this shift. Adopting the standard assumption that the activation function $\sigma(\cdot)$ (e.g., ReLU, Leaky ReLU) satisfies Lipschitz continuity with Lipschitz

constant $K = 1$, we bound the feature deviation as:

$$\|\Delta \mathbf{x}^{l+1}\|_2 = \|\sigma((\mathbf{W}_{t-1}^l + \Delta \mathbf{W}_t^l) \mathbf{x}^l) - \sigma(\mathbf{W}_{t-1}^l \mathbf{x}^l)\|_2 \leq \|\Delta \mathbf{W}_t^l \mathbf{x}^l\|_2. \quad (3)$$

We define the **feature drift risk** ($\mathcal{E}_{\text{feat}}$) as the expected squared change over the historical data distribution $\mathcal{D}_{1:t-1} = \bigcup_{i=1}^{t-1} \mathcal{D}_i$. Using the linearity of expectation and the cyclic property of the trace operator, the upper bound of the stability risk is derived as:

$$\mathcal{E}_{\text{feat}} = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{1:t-1}} [\|\Delta \mathbf{x}^{l+1}\|_2^2] \leq \text{Tr}((\Delta \mathbf{W}_t^l)^\top (\Delta \mathbf{W}_t^l) \Sigma_{1:t-1}^l), \quad (4)$$

where $\Sigma_{1:t-1}^l = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{1:t-1}} [\mathbf{x}^l (\mathbf{x}^l)^\top]$ denotes the uncentered covariance matrix of the input features at the l -th layer from stages 1 to $t-1$. We perform spectral decomposition on this covariance matrix. Let

$$\Sigma_{1:t-1}^l = \sum_{k=1}^{d_{in}} \lambda_k \mathbf{u}_k \mathbf{u}_k^\top, \quad (5)$$

where $\{\mathbf{u}_k, \lambda_k\}$ denote the normalized eigenvector-eigenvalue pairs. Substituting this into Eq. (19) yields a decoupled upper bound:

$$\mathcal{E}_{\text{feat}} \leq \text{Tr} \left((\Delta \mathbf{W}_t^l)^\top (\Delta \mathbf{W}_t^l) \sum_{k=1}^{d_{in}} \lambda_k \mathbf{u}_k \mathbf{u}_k^\top \right) = \sum_{k=1}^{d_{in}} \lambda_k \|\Delta \mathbf{W}_t^l \mathbf{u}_k\|_2^2. \quad (6)$$

This formulation reveals that the feature drift risk of a parameter update direction is proportional to its corresponding eigenvalue in the historical feature covariance matrix. Consequently, we design a regularization term scaled by the eigenvalues of the parameters to minimize feature drift.

4 Methodology

4.1 Eigenvalue Scaled Projection Regularization

Motivated by the theoretical analysis in Sec. 3.3, we propose the Eigenvalue Scaled Projection Regularization method to demystifying feature drift of YOLO-based incremental detectors. Consider the incremental learning step from task t to $t+1$. We define the parameter deviation of the l -th layer as the difference between the current weights and the consolidated weights from the previous stage:

$$\Delta \mathbf{W}_{t+1}^l = \mathbf{W}_{t+1}^l - \mathbf{W}_t^l. \quad (7)$$

To control the feature drift caused by this deviation, we utilize the feature drift risk upper bound derived in Eq. (21) as regularization objective. Directly using raw eigenvalues λ_k as regularization coefficients can lead to optimization difficulties, because the eigenvalues of significant directions change vastly (as is shown in Fig. 2b). To prevent the penalty on the most significant direction dominating the regularization, we introduce a calibrated importance score λ_k^* . Let $0 \leq r \leq d_{in}$ be the elbow point of eigenvalue sequence, where the second order difference reach the maximum value (refer [30]). The calibrated score is defined as:

$$\lambda_k^* = \min \left(\frac{\lambda_k}{\lambda_r}, 1 \right). \quad (8)$$

This formulation creates a truncated penalty weight: for dominant principal components ($\lambda_k > \lambda_r$), the penalty saturates at 1 and contributes to the regularization evenly, while for tail components ($\lambda_k < \lambda_r$), the penalty decays linearly with their importance.

Based on the feature drift risk upper bound derived in Eq. (21), we formulate the Eigenvalue Scaled Projection Regularization (ESPReg) loss as:

$$\mathcal{L}_{\text{reg}} = \frac{1}{L} \sum_{l=1}^L \sum_{k=1}^{d_{in}} \lambda_k^* \|\Delta \mathbf{W}_t^l \mathbf{u}_k\|_2^2. \quad (9)$$

Here, the term $\|\Delta \mathbf{W}_t^l \mathbf{u}_k\|_2^2$ quantifies the projection of the weight update onto the k -th principal component of historical features. By weighting this projection with λ_k^* , ESPReg effectively reshapes the optimization landscape: it imposes stringent regularization penalties along principal directions crucial for historical tasks (where λ_k^* is large), while permitting unconstrained parameter updates within the null space (where $\lambda_k^* \approx 0$).

Finally, the total objective function for the current task combines the standard detection loss with the proposed regularization:

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \alpha \mathcal{L}_{\text{reg}}, \quad (10)$$

where $\mathcal{L}_{\text{det}}$ includes classification and regression losses for the new task, and α is a hyperparameter controlling the overall strength of stability preservation.

4.2 Confidence-Aware Distillation

As empirical results in Fig. 1b, **ESPReg** alone cannot completely mitigate the severe cross-task class confusion.

To resolve this issue, we introduce a Confidence-Aware Distillation (CAD) method. Let $\mathcal{M}_t$ represent the frozen base model from the previous stage and $\mathcal{M}_{t+1}$ denote the current model being trained. For a given input image, the YOLO detection head outputs predictions for N anchors. Let $\mathbf{p}_{t,i}$ and $\mathbf{p}_{t+1,i}$ denote the classification probability vectors for the i -th anchor generated by the base and current models, respectively.

Instead of indiscriminately distilling all output channels across all anchors, which inevitably includes task-irrelevant background noise and suppresses the learning of novel classes, we selectively distill knowledge guided by the base model’s confidence. Specifically, for the i -th anchor, we calculate the maximum classification score across all previous class channels $\mathcal{C}_t$. We define this maximum score as the confidence mask m_i :

$$m_i = \max_{c \in \mathcal{C}_t} p_{t,i,c}. \quad (11)$$

Let $\hat{c}_i$ be the specific category index where this maximum score is achieved:

$$\hat{c}_i = \arg \max_{c \in \mathcal{C}_t} p_{t,i,c}. \quad (12)$$

To effectively transfer the classification knowledge while avoiding interference with new classes, we compute the Kullback-Leibler (KL) divergence $[\]$ between the base and current models exclusively on this most prominent channel $\hat{c}_i$. By employing m_i as a weighting factor, we compel the optimizer to strictly anchor its predictions on regions where the base model exhibits high confidence, while granting plasticity to ambiguous or background anchors. The proposed classification distillation loss is formulated as a weighted average:

$$\mathcal{L}_{\text{cad}} = \frac{1}{\sum_{j=1}^N m_j} \sum_{i=1}^N m_i \cdot \mathcal{D}_{\text{KL}}(p_{t,i,\hat{c}_i} \parallel p_{t+1,i,\hat{c}_i}), \quad (13)$$

where $\mathcal{D}_{\text{KL}}(p \parallel q) = p \log(\frac{p}{q}) + (1-p) \log(\frac{1-p}{1-q})$ denotes the binary KL divergence.

Consequently, the overall objective function presented in Eq. (10) is updated to incorporate this distillation strategy:

$$\mathcal{L} = \mathcal{L}_{\text{det}} + \alpha \mathcal{L}_{\text{reg}} + \beta \mathcal{L}_{\text{distill}}, \quad (14)$$

where β is a hyperparameter balancing the distillation strength. This dual-regularization approach ensures that feature drift is minimized via ESPReg while cross-task class confusion is demystified via Confidence-Aware Distillation.

5 Experiments

Datasets and Protocols. Our experiments are conducted on Pascal VOC [3, 4] and MS COCO 2017 [13] datasets, which are widely adopted benchmarks for IOD. We comprehensively evaluate the performance of our ESP-YOLO on both single-step incremental settings and multi-step incremental settings. Specifically, for Pascal VOC, we adopt three single-step splits (19+1 and 15+5) and three multi-step splits (10-5 with 3 tasks, 5-5 with 4 tasks, and 10-2 with 6 tasks); for MS COCO 2017, we carry out evaluation on two standard single-step splits (70+10 and 40+40), covering different incremental learning difficulty levels. In the $A+B$ or $A-B$ protocol, the detector is first trained on A base classes and then sequentially updated with B novel classes at each incremental stage. Following prior IOD protocols [14, 30], we report *Old*, *New*, *All*, and *Avg* on Pascal VOC, denoting mAP over old classes, new classes, all seen classes, and the average of Old and New, respectively.

Implementation Details. We adopt YOLOv8 [26] as the base detector, upon which we apply ESPReg and CAD to build our ESP-YOLO. The model is trained for 100 epochs per incremental

Table 1: **Comparison with state-of-the-art IOD methods on Pascal VOC test 2007.** We highlight the **best** and second-best results. '-' indicates results not available. Note that Joint and Fine-tuning serve as the theoretical upper and lower bounds for the results, and are therefore excluded from the ranking.

Method	19+1				15+5			
	Old	New	All	Avg	Old	New	All	Avg
Joint	78.1	82.8	78.0	80.5	79.8	79.1	78.0	79.5
Fine-tuning	0.0	82.4	3.9	41.2	0.0	78.4	11.5	39.2
FasterILOD [20]	68.9	61.1	68.5	65.0	71.6	56.9	67.9	64.3
ILOD-Meta [9]	70.9	57.6	70.2	64.2	71.7	55.9	67.8	63.8
ORE [8]	69.4	60.1	68.9	64.7	71.8	58.7	68.5	65.2
OW-DETR [6]	70.2	62.0	69.8	66.1	72.2	59.8	69.1	66.0
PPAS [37]	70.5	53.0	69.2	61.8	-	-	-	-
MVC [34]	70.2	60.6	69.7	65.4	69.4	57.9	66.5	63.7
PROB [39]	73.9	48.5	72.6	61.5	73.5	60.8	70.1	67.0
PseudoRM [33]	72.9	67.3	72.6	70.1	73.4	60.9	70.3	66.9
MMA [1]	71.1	63.4	70.7	67.2	73.0	60.5	69.9	66.7
BPF [17]	74.5	65.3	74.1	69.9	75.9	63.0	72.7	69.5
OSR [31]	71.8	66.5	71.5	69.2	72.2	61.2	69.5	66.7
ABR [14]	71.0	<u>69.7</u>	70.9	70.4	73.0	<u>65.1</u>	71.0	69.1
NSGP-RePRE [30]	<u>76.3</u>	69.0	76.0	<u>72.7</u>	<u>77.5</u>	61.8	<u>73.6</u>	<u>69.7</u>
ESP-YOLO	76.9	72.0	<u>75.5</u>	74.5	80.4	77.1	75.0	78.8

Table 2: **Comparison with state-of-the-art IOD methods on the MS COCO 2017 validation set.** All metrics are computed over all seen classes after the single-step incremental stage. We highlight the **best** and second-best results among incremental methods. '-' indicates unavailable results. Joint training serves as the reference upper bound and is excluded from the ranking.

Method	40+40 Task			70+10 Task		
	mAP	AP75	AP50	mAP	AP75	AP50
Joint	53.9	58.8	70.4	53.9	58.8	70.4
ILOD-Meta	23.8	24.4	40.5	-	-	-
ABR	34.5	35.2	<u>57.8</u>	31.1	32.7	52.9
FasterILOD	20.6	-	40.1	21.3	-	39.9
PseudoRM	25.3	-	44.4	-	-	-
MMA	33.0	34.6	56.6	30.2	31.5	52.1
BPF	34.4	37.3	54.3	36.2	38.9	<u>56.8</u>
NSGP [30]	<u>35.4</u>	<u>38.6</u>	55.3	<u>36.5</u>	<u>39.8</u>	56.0
ESP-YOLO	42.5	46.4	58.0	44.8	48.7	60.6

stage with a batch size of 16, using the SGD optimizer with a learning rate of 0.01, momentum of 0.937, and weight decay of 5×10^{-4} . All experiments are performed on a single NVIDIA 4090 GPU. The hyper-parameter of α and β in 14 are both 100. For fair comparison, we report published results for existing IOD methods when available and re-implement two standard reference bounds following prior protocols [14, 17, 30]: *Joint*, trained on the union of all stage-wise data with complete annotations, and *Fine-tuning*, sequentially trained on incremental data without replay, distillation, or regularization.

Comparison Results. Tabs. 1, 2, 3 summarize the comparison results between our proposed ESP-YOLO and comparable IOD models on different benchmarks. On the Pascal VOC single-step incremental benchmark (Tab. 1), our ESP-YOLO achieves significant performance gains and comprehensive superiority over existing SOTA methods across most evaluation metrics and settings. For the 19+1 and 15+5 settings, ESP-YOLO achieves the best performance on Old class mAP, New class mAP and average mAP (Avg). Specifically, in the 19+1 setting, our method reaches 76.9% Old class mAP, 72.0% New class mAP and 74.5% Avg mAP, all ranking first among all compared methods; in the more challenging 15+5 setting, ESP-YOLO further expands its advantages, achieving 80.4% Old class mAP, 77.1% New class mAP, 75.0% All class mAP and 78.8% Avg mAP, all of which are the best results in the comparison.

Table 3: **Comparison with state-of-the-art IOD methods on Pascal VOC test 2007 in multi-step settings.** For each protocol, the three columns report mAP on old/base classes, newly introduced classes, and all seen classes after completing all incremental stages, respectively. We highlight the **best** and **second-best** results among incremental methods. '-' indicates results not available. Note that Joint and Fine-tuning serve as the theoretical upper and lower bounds for the results, and are therefore excluded from the ranking.

Method	10-5 (3 tasks)			5-5 (4 tasks)			10-2 (6 tasks)		
	1-10	11-20	1-20	1-5	6-20	1-20	1-10	11-20	1-20
Joint	80.5	79.7	78.0	78.7	79.0	78.0	80.5	79.7	78.0
Fine-tuning	5.3	30.6	18.0	0.5	18.3	13.8	3.8	13.6	8.7
ABR (Yuyang et al., 2023)	68.7	67.1	67.9	64.7	56.4	58.4	67.0	58.1	62.6
FasterILOD (Ren et al., 2016)	68.3	57.9	63.1	55.7	16.0	25.9	64.2	48.6	56.4
MMA (Cermelli et al., 2022)	66.7	61.8	64.2	62.3	31.2	38.9	65.0	53.1	59.1
BPF (Mo et al., 2024)	69.1	68.2	68.7	60.6	63.1	62.5	68.7	56.3	62.5
NSGP-RePRE	72.4	67.6	70.0	64.6	66.1	65.7	70.1	58.8	64.4
ESP-YOLO	74.2	69.4	70.8	70.1	67.3	66.2	69.7	77.9	71.6

Table 4: **Ablation study of different components on Pascal VOC test 2007 (15+5 setting).** All experiments are built upon the YOLOv8 (Base). PL: Pseudo Label, ESPReg: Eigenvalue Scaled Projection Regularization, CAD: Confidence-Aware Distillation.

Base	PL	ESPReg	CAD	15+5			
				Old	New	All	Avg
✓				0.0	79.0	11.0	39.5
✓	✓			51.0	78.3	49.0	64.7
✓	✓	✓		72.6	76.8	66.6	74.7
✓	✓		✓	65.3	76.9	67.4	71.1
✓	✓	✓	✓	80.4	77.1	75.0	78.8

On MS COCO 2017 single-step incremental benchmark (Tab. 2), ESP-YOLO also exhibits prominent performance advantages, outperforming all compared methods on all evaluation metrics for both 40+40 and 70+10 settings. In the 40+40 task, ESP-YOLO achieves 42.5% mAP, 46.4% AP75 and 58.0% AP50, surpassing the previous best method by a large margin; in the 70+10 task, our method further reaches 44.8% mAP, 48.7% AP75 and 60.6% AP50, consistently ranking first in all metrics. These results fully validate the strong generalization ability and effectiveness of our ESP-YOLO on large-scale datasets with more complex categories.

For the multi-step incremental learning settings on Pascal VOC (Tab. 3), which are closer to real-world continual learning scenarios, ESP-YOLO also outperforms existing methods. In all three multi-task incremental settings (10-5 with 3 tasks, 5-5 with 4 tasks, 10-2 with 6 tasks), the results of ESP-YOLO are among the top tier, and show promising performance in both old knowledge retention and new class learning.

6 Ablations

Our ablation experiments are conducted on the Pascal VOC 15+5 single-stage class-incremental detection setting to evaluate the contribution of each proposed component.

6.1 Effectiveness of Components

We analyze the impact of ESPReg and CAD on the incremental learning performance in Tab.4. Here, *Base* denotes the YOLOv8 under sequential fine-tuning, while *PL* denotes pseudo labeling [14, 25], which is a common method to preserve old-class supervision. Several key observations can be drawn: (1) **ESPReg significantly mitigates forgetting**: Compared to the baseline (Base in Tab.4) with only Pseudo Labeling [17] (PL in Tab.4), adding ESPReg improves the mAP on old classes (Task-1) from 51.0% to 72.6%, demonstrating its superior capability in preserving historical knowledge. (2) **CAD enhances task-agnostic performance**: While ESPReg focuses on stability, CAD bridges the performance gap between task-aware and task-agnostic scenarios by eliminating cross-task class confusion, which is further demonstrated in Sec. 6.3. (3) **Synergistic Effect**: The combination of both components achieves the optimal incremental result (78.8% Avg mAP).

6.2 Effectiveness of ESPReg

To quantitatively assess the stability of the representation, we first measure the **feature drift** of the feature extractor after incremental training. Specifically, we randomly select 100 samples from the

Table 5: **Quantitative analysis of feature drift.** We report the average Euclidean distance of backbone features between the base model and the model after Task-2 training (15+5 setting).

Method	Fine-tuning	EWC	NSGP	ESPReg
Feature Drift ($\downarrow$)	13.12	11.17	10.25	6.84

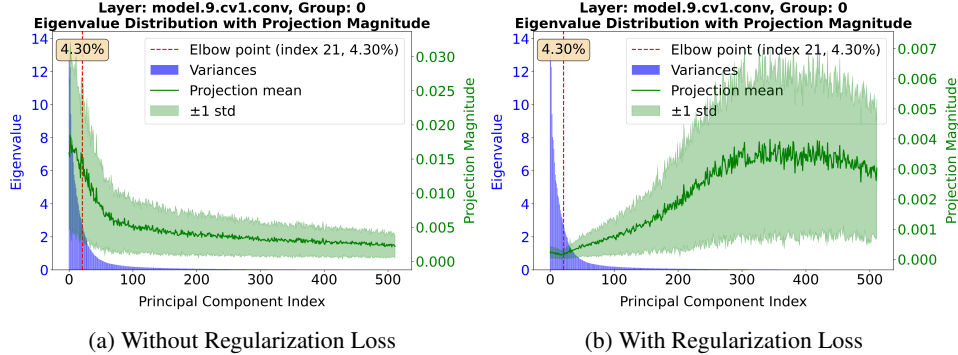


Figure 2: **Effectiveness of ESPReg.** We visualize the projection magnitude of weight updates (green curve) against the historical feature eigenvalues (blue bars). The elbow point is also represented (red dotted line). (a) Without ESPReg, the weight updates dominate the significant subspace, leading to drastic intermediate feature drift. (b) With ESPReg, weight updates are explicitly penalized in the significant subspace (left) and concentrated in the null space (right), indicating minimal intermediate feature drift.

303 historical tasks and extract their features using both the initial base model and the incrementally
 304 updated model. We then compute the average Euclidean distance between these two sets of features.
 305 As shown in Tab. 5, the model regularized by ESPReg exhibits significantly lower feature drift
 306 compared to the vanilla finetuning model, the model regularized by EWC [10] and NSGP [30]. And
 307 as is shown in Fig. 1a, the feature drift of YOLO+ESPReg is significantly lower than Faster R-CNN.
 308 These results confirm that ESPReg effectively stabilizes the intermediate feature of IOD model.

309 We then visualize the parameter changes ΔW in the first convolutional block of layer 9 (last layer
 310 of feature extractor). Specifically, we plot the magnitude of these weight updates projected onto
 311 the eigenvectors of the historical feature covariance, overlaid with the eigenvalue distribution. As
 312 illustrated in Fig. 2a, without regularization, the optimization naturally gravitates towards the **signifi-**
 313 **cant subspace** (the left region with high eigenvalues), causing severe interference with historical
 314 knowledge. In contrast, Fig. 2b demonstrates that ESPReg effectively **suppresses updates in these**
 315 **high-variance directions**. Instead, it steers the optimization towards the **null space** (the right region
 316 beyond the red dotted line), thereby preserving stability while utilizing the low-risk subspace for
 317 plasticity. The red dotted line represents elbow point defined by [30].

318 6.3 Effectiveness of CAD

319 We further investigate the cross-task confusion using confusion matrices (visualized in Fig. 1). As
 320 shown in Fig. 1b, when only ESPReg is employed to mitigate forgetting, the model’s classifier remains
 321 susceptible to class imbalance within the training set. This leads to Task-1 categories being frequently
 322 misclassified as visually similar categories in Task-2, causing a noticeable drop in task-agnostic
 323 detection accuracy. As shown in Fig. 1c, the introduction of the CAD method significantly alleviates
 324 this cross-task confusion. The lower-left region of the confusion matrix becomes markedly sparser.
 325 For instance, in the challenging "cow-sheep" confusion group, the misclassification rate decreases
 326 from 73% to 42%. Consequently, the task-agnostic detection accuracy with CAD approaches the
 327 median of task-aware detection accuracies across individual tasks.

328 7 Conclusion and Discussion

329 In this paper, we propose Eigenvalue Scaled Projection Regularization (ESPReg) to resolve the
 330 drastic feature drift and cross-task class confusion in YOLO-based Incremental Object Detector.
 331 Grounded in a rigorous theoretical analysis of the feature drift risk, ESPReg provides an adaptive,
 332 eigenvalue-weighted regularization to dynamically balance the retention of historical knowledge with
 333 the learning of new concepts. Additionally, our Confidence-Aware Distillation demystifies cross-task
 334 class confusion, ensuring performance in task-agnostic inference scenario. Extensive experiments on
 335 Pascal VOC and MS COCO confirm that ESPReg consistently outperforms state-of-the-art methods.

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447 A Technical appendices and supplementary material

448 A.1 Disparity of Feature Drift between YOLO and Faster R-CNN

449 Given the architectural heterogeneity between YOLO and Faster R-CNN frameworks, a direct layer-
 450 wise comparison of their internal feature drift is intractable. Consequently, we employ the feature
 451 drift at the output layer of the feature extractor—specifically, the layer immediately preceding the
 452 classification and regression heads—as a representative metric for our analysis.

453 Formally, the feature drift $\mathcal{E}_{feat}^{L_f}$ is calculated as follows:

$$\mathcal{E}_{feat}^{L_f} = \frac{1}{HW} \sum_{i=1}^W \sum_{j=1}^H \left\| \mathbf{F}_{t+1}^{L_f}[i, j] - \mathbf{F}_t^{L_f}[i, j] \right\|_2, \quad (15)$$

454 where L_f denotes the index of the final feature extraction layer before the prediction heads. $\mathbf{F}_t^{L_f}$ and
 455 $\mathbf{F}_{t+1}^{L_f}$ represent the output feature maps at layer L_f yielded by the detector after training on task t and
 456 task $t + 1$, respectively. Furthermore, $\|\cdot\|_2$ denotes the L_2 -norm, while H and W correspond to the
 457 spatial height and width of the feature map.

458 A.2 Detailed Analysis of Feature Drift

459 To establish a theoretical bound on the feature drift of YOLO-based incremental detectors, we analyze
 460 the influence of parameter updates on the network’s intermediate features. Without loss of generality,
 461 consider the l -th layer of the network (e.g., fully connected or convolutional) with input features
 462 $\mathbf{x}^l \in \mathbb{R}^{d_{in}}$. Let $\mathbf{W}_{t-1}^l \in \mathbb{R}^{d_{out} \times d_{in}}$ denote the parameters converged at stage $t - 1$. The forward
 463 propagation is given by:

$$\mathbf{x}^{l+1} = \sigma(\mathbf{W}_{t-1}^l \mathbf{x}^l), \quad (16)$$

464 where $\sigma(\cdot)$ is a non-linear activation function. When learning the new task at stage t , the weights are
 465 updated by a perturbation $\Delta \mathbf{W}_t^l$, resulting in $\mathbf{W}_t^l = \mathbf{W}_{t-1}^l + \Delta \mathbf{W}_t^l$. The resulting shift in the output
 466 feature is:

$$\Delta \mathbf{x}^{l+1} = \sigma((\mathbf{W}_{t-1}^l + \Delta \mathbf{W}_t^l) \mathbf{x}^l) - \sigma(\mathbf{W}_{t-1}^l \mathbf{x}^l). \quad (17)$$

467 To quantify forgetting, we examine the magnitude of this shift. Adopting the standard assumption that
 468 the activation function $\sigma(\cdot)$ (e.g., ReLU, Leaky ReLU) satisfies Lipschitz continuity with Lipschitz
 469 constant $K = 1$, we bound the feature deviation as:

$$\begin{aligned} \|\Delta \mathbf{x}^{l+1}\|_2 &= \|\sigma((\mathbf{W}_{t-1}^l + \Delta \mathbf{W}_t^l) \mathbf{x}^l) - \sigma(\mathbf{W}_{t-1}^l \mathbf{x}^l)\|_2 \\ &\leq \|(\mathbf{W}_{t-1}^l + \Delta \mathbf{W}_t^l) \mathbf{x}^l - \mathbf{W}_{t-1}^l \mathbf{x}^l\|_2 \\ &= \|\Delta \mathbf{W}_t^l \mathbf{x}^l\|_2. \end{aligned} \quad (18)$$

470 We define the **feature drift risk** ($\mathcal{E}_{feat}$) as the expected squared change over the historical data
 471 distribution $\mathcal{D}_{1:t-1} = \bigcup_{i=1}^{t-1} \mathcal{D}_i$. Using the linearity of expectation and the cyclic property of the trace
 472 operator, the upper bound of the stability risk is derived as:

$$\begin{aligned} \mathcal{E}_{feat} &= \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{1:t-1}} [\|\Delta \mathbf{x}^{l+1}\|_2^2] \\ &\leq \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{1:t-1}} [\|\Delta \mathbf{W}_t^l \mathbf{x}^l\|_2^2] \\ &= \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{1:t-1}} [(\mathbf{x}^l)^\top (\Delta \mathbf{W}_t^l)^\top (\Delta \mathbf{W}_t^l) \mathbf{x}^l] \\ &= \text{Tr}((\Delta \mathbf{W}_t^l)^\top (\Delta \mathbf{W}_t^l) \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{1:t-1}} [\mathbf{x}^l (\mathbf{x}^l)^\top]) \\ &= \text{Tr}((\Delta \mathbf{W}_t^l)^\top (\Delta \mathbf{W}_t^l) \Sigma_{1:t-1}^l), \end{aligned} \quad (19)$$

473 where $\Sigma_{1:t-1}^l = \mathbb{E}_{\mathbf{x} \sim \mathcal{D}_{1:t-1}} [\mathbf{x}^l (\mathbf{x}^l)^\top]$ denotes the uncentered covariance matrix of the input features
 474 at the l -th layer from stages 1 to $t - 1$. We perform spectral decomposition on this covariance matrix.
 475 Let

$$\Sigma_{1:t-1}^l = \sum_{k=1}^{d_{in}} \lambda_k \mathbf{u}_k \mathbf{u}_k^\top, \quad (20)$$

476 where $\{\mathbf{u}_k, \lambda_k\}$ denote the normalized eigenvector-eigenvalue pairs. Substituting this into Eq. (19)
 477 yields a decoupled upper bound:

$$\begin{aligned}
 \mathcal{E}_{\text{feat}} &\leq \text{Tr} \left((\Delta \mathbf{W}_t^l)^\top (\Delta \mathbf{W}_t^l) \sum_{k=1}^{d_{in}} \lambda_k \mathbf{u}_k \mathbf{u}_k^\top \right) \\
 &= \sum_{k=1}^{d_{in}} \lambda_k \cdot \text{Tr} \left((\Delta \mathbf{W}_t^l)^\top (\Delta \mathbf{W}_t^l) \mathbf{u}_k \mathbf{u}_k^\top \right) \\
 &= \sum_{k=1}^{d_{in}} \lambda_k \left\| \Delta \mathbf{W}_t^l \mathbf{u}_k \right\|_2^2.
 \end{aligned} \tag{21}$$

478 This formulation reveals that the feature drift risk of a parameter update direction is proportional to
 479 its corresponding eigenvalue in the historical feature covariance matrix. Consequently, we design a
 480 regularization term scaled by the corresponding eigenvalues of the parameters to minimize feature
 481 drift.

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